

Dual-stream multimodal shared prompt model for fake news detection

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ABSTRACT

Current mainstream fake news detection methods primarily rely on pretrained models for encoding and feature extraction of both images and text, and then design interactive network modules based on the extracted features to achieve multimodal feature interaction. However, these interactions are often shallow and may not explore the complex relationships between image and text information, nor fully harness the potential of the advanced architecture of pretrained models. In response to these challenges, we propose a dual-stream multimodal shared prompt (DMSP) method based on pretrained models, which exploits the complementarity between image and text by introducing a learnable multimodal shared prompt into the image-text encoder. This enables the model to generate dual-stream deep image-text feature interaction vectors, maximizing the deep correlations and complementarity between image and text during the multimodal information interaction process. Furthermore, we design a pseudo-siamese attention network that learns multimodal interaction feature from two perspectives, local features and global features. This network effectively captures important interaction features, thereby enhancing the model's ability to identify fake news. Finally, a classifier is used to perform binary classification on the dual-perspective deep image-text mixed features. Experimental results show that DMSP outperforms state-of-the-art baseline models with only five training epochs, achieving a 3.7% and 1% improvement in detection accuracy in the public PolitiFact and GossipCop datasets, respectively.

1. Introduction

In today's digital information era, multimedia social platforms have become the primary channels for obtaining external news and social interactions. However, with the rapid growth of social media users and the lack of effective fake news regulation tools, the issue of fake news dissemination has become increasingly severe (Zhang et al., 2025; Zhou et al., 2020). Fake news not only negatively affects people's daily lives, but is also often maliciously distorted and misleading in reports on major events (Peng et al., 2024; Zhou et al., 2023). During the COVID-19 pandemic, all social media platforms are flooded with fake news about virus transmission and treatment options (Biamby et al., 2022). During election periods, rumors and fabricated stories about candidates emerged incessantly, severely misleading voters' judgments (Nielsen & McConville, 2022). Furthermore, fake news detection plays a fundamental and

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central role in the entire misinformation governance process, and it is particularly indispensable for key downstream tasks like source detection (Carragher et al., 2024; Hou et al., 2024). However, achieving source detection critically depends on the accurate identification of fake news. If fake news detection is inaccurate, source tracing efforts may be misled, rendering intervention strategies ineffective (Sivasankari & Vadivu, 2022). Therefore, improving the accuracy of fake news detection is the foundation for supporting subsequent tasks, and it has quickly become a research focus due to its key role in preventing misinformation.

With continuous updates and iterations of social platforms, the form of fake news has also changed significantly. Initially, social platforms mainly published tweets in text form, and readers obtained information through textual content (Jing et al., 2023; Wu et al., 2023b). However, with the popularization of multimedia content on social media, fake news has gradually evolved to be spread through a multimodal combination of text and images. Jiang et al. (2024). Early unimodal methods focused mainly on text analysis. These methods rely on natural language processing (NLP) techniques to analyze linguistic patterns, sentiment tendencies, word distributions, and other features of articles to determine the authenticity of the news (Allein et al., 2021). Despite the success of unimodal approaches, the increasing complexity of fake news forms make it difficult to cope with the challenges of multimodal news. Thus, fake news detection methods have gradually shifted towards multimodal approaches (Li et al., 2024). These multimodal methods assess the authenticity of news by jointly processing visual and textual information (Chen et al., 2022). They typically employ pretrained models such as RoBERTa, VGG, and CLIP to extract text and image features (Chen et al., 2022; Yang et al., 2024a). Subsequently, they establish semantic associations between textual and visual features, achieving a comprehensive judgment of news authenticity. In general, as news forms become increasingly diverse, multimodal methods demonstrate greater adaptability by integrating textual and visual information, gradually becoming a key focus in current research.

However, current research generally adopts shallow fusion strategies for image-text interaction, such as feature concatenation, simple stacking, or lightweight attention mechanisms, resulting in limited and shallow interactions (Shan et al., 2024; Wang & Zhang, 2024). These approaches lack the capability to effectively capture deep and complex correlations between images and text, making it difficult to model the intricate dependencies often hidden within fake news content. In terms of using pretrained models, existing methods typically employ pretrained visual and language models as image and text feature extractors, instead of leveraging advanced architectural advantages to achieve deep multimodal interaction (Mostafa et al., 2024), which limits the model's ability to align features and express fused multimodal representations. Moreover, current approaches often pass the fused image-text information directly to the classifier, or only focus on global feature extraction (Li et al., 2021), thus overlooking the crucial role of local details in the detection task.

In this work, we believe the pretrained models should not only be considered as encoders and feature extractors, but also their generalization and detection capabilities should be more deeply integrated into the detection models. Therefore, we propose a dual-stream multimodal shared prompt model based on pretrained models, aiming to fully leverage the advanced architectural advantages and generalization capacity of pretrained models. And we also propose a dual-perspective feature learning approach, which performs parallel learning of multimodal fused information from two complementary perspectives (global and local). This approach is able to comprehensively mine useful information at different levels within multimodal fused data, providing more accurate and robust multimodal features. Specifically, without altering the original architecture of the pretrained encoders, we introduce shared image prompts and text prompts into the pretrained text encoder and pretrained image encoder, respectively. This process generates deep interactive feedback between text and image information. The obtained feedback is then processed through a pseudo-siamese attention network, which learns associated features from both local and global perspectives. Finally, the results of dual-perspective learning are jointly evaluated to predict the authenticity of the news.

Our main contributions are as follows.

- This is the first multimodal shared prompt learning method applied to fake news detection. By introducing a learnable multimodal shared prompt into the image-text encoder, our model can better leverage the advanced architecture of pretrained models. This design enables the model to capture deeper correlations between images and text, addressing the limitation of existing methods that do not fully harness the potential of pretrained model architectures.
- We propose a dual-perspective pseudo-siamese attention network. It enables parallel learning of global and local perspectives when processing deep image-text mixed information. This effectively captures important coarse-grained and fine-grained interactive features, further enhancing the ability to identify fake news.
- The experimental results show that our method outperforms the state-of-the-art baseline models with five training epochs. Specifically, the detection accuracy on the public PolitiFact and GossipCop datasets improved by 3.7% and 1%, respectively.

The rest structure of this paper is as follows. Section 2 provides a brief overview of the related work on both unimodal and multimodal fake news detection. Section 3 outlines the research objectives. Section 4 describes our DMSP method. Section 5 presents and analyzes the experimental results. Finally, Section 6 concludes this work.

2. Related work

Fake news appears across many areas of social life, with major categories including politics, health, entertainment, finance, society, education, disasters, military affairs, technology, and conflicts (Nan et al., 2021; Zhou et al., 2024). Different categories serve different purposes and exhibit different informational characteristics. For example, health-related fake news (such as misinformation about COVID-19) may mislead the public about disease risks and prevention policies, creating unnecessary panic. Political fake news may

manipulate public attitudes toward political events or figures, affecting public opinion and political trust. Entertainment fake news often exaggerates or fabricates celebrity events to draw attention, causing emotional reactions. Financial fake news may mislead investors and result in economic losses. These categories differ in writing style, emotional tone, as well as in visual presentation style, target audience. Such diversity can strongly influence user perception and behavior. It also brings challenges for detection models, which may face semantic shifts or domain discrepancies when handling cross-category fake news. To effectively identify these diverse fake news stories, numerous fake news detection methods have been proposed. Existing methods can be categorized into unimodal and multimodal methods based on the modal type used. A brief review of these two types of methods is provided below.

2.1. Unimodal fake news detection

In the early stages, researchers employed manually designed statistical rules to analyze inherent features, such as word frequency, the usage of content and function words, punctuation marks, etc., and used these statistical features for fake news detection (Wang et al., 2018). However, as the content of fake news became increasingly complex, methods based on manually designed statistical rules gradually proved to be insufficient. Later, researchers began to apply NLP techniques in fake news detection, replacing the traditional method of manually designing statistical rules. This shift significantly reduced the actual workload, as NLP techniques can automatically extract and process important features from large volumes of textual data, minimizing the need for human intervention. At the same time, this approach increased the flexibility and adaptability of the detection models (Nan et al., 2021). With further research, scholars discovered significant differences in emotional stances between real and fake news, leading to the introduction of sentiment analysis techniques (Chakraborty et al., 2023). This approach analyzes the emotional tone of the text and incorporates it as an important feature. Additionally, user comments and replies are recognized as significant corroborative features for identifying fake news (Wang et al., 2024a), as these interactions can provide evidence and support for authenticity assessment, thereby improving detection accuracy. Based on this, a dual sentiment detection technique that combines sentiment analysis and comment response analysis is proposed (Zhang et al., 2021). This technique learns the text content and emotional representations of both the publisher and the commentators, utilizing sentiment disparity analysis to identify fake news, highlighting the crucial role of sentiment analysis. With the rapid development of pretraining technology, detection methods based on pretrained models demonstrate clear advantages (Mostafa et al., 2024). Researchers propose propaganda detection methods based on pretrained models, aiming to uncover the mechanism of propaganda infiltration into news content (Ahmad et al., 2025a). Some studies use pretrained large language models (LLMs) not only to extract and jointly model complex signals (Leite et al., 2024), but also to rewrite and verify news content (Wu et al., 2024). In addition, some approaches focus on content verification and source identification, further promoting the application of pretrained models in fake news detection (Ahmad et al., 2025b).

However, when facing multimodal fake news, these methods struggle to address the challenges posed by the complex and diverse nature of fake news. This is mainly because unimodal approaches cannot process the different modalities in multimodal fake news, nor can they capture the crucial associations between modalities. It is difficult to detect multimodal fake news using only textual or visual features. Therefore, multimodal detection methods have become the primary focus of current research.

2.2. Multimodal fake news detection

Multimodal fake news detection methods combine visual and textual information from the news, focusing on learning the interaction relationships between different modalities. The results of the multimodal interaction are then used as the key basis for detecting fake news. In early research, scholars mainly used convolutional networks to learn and extract features from different modalities (Jamshidi et al., 2024; Zhang et al., 2022). These features are then interactively combined through simple concatenation, and the combined features are used as key evidence to determine the authenticity of the news (Singhal et al., 2022). With the rapid evolution and increasing complexity of multimodal fake news content, simple feature concatenation is no longer effective in capturing robust interactions between text and images. The researchers shifted their focus to extracting stable and meaningful correlations between text and image (Wu et al., 2023a, 2021). With further research, it was discovered that encoding-decoding approaches, such as variational encoders (Khattar et al., 2019), could effectively extract complex text-image relationships by assigning different weights to different features before fusion. However, these methods focused primarily on global-level fusion, neglecting the importance of local feature information. To address this, researchers began to focus on entity-oriented multigranularity feature alignment methods, which can match image and text features from multiple granularities (Li et al., 2021). In order to introduce more corroborated information, the concept of external knowledge is incorporated into fake news detection methods (Hu et al., 2021; Wang et al., 2021). By comparing the news with external knowledge, the accuracy and interpretability of the models are further enhanced. Recent research has started to employ visual-language models (VLMs) (Xu et al., 2024a; Zhou et al., 2023) and LLMs for fake news detection task (Xu et al., 2024b; Yi et al., 2025). LLMs, with their powerful contextual understanding of text, are used to process news articles, while VLMs benefit from their image-text encoders to reduce the distribution gap between visual and textual, improving detection accuracy. As pretrained models become more powerful, prompt learning is introduced into fake news detection. It helps improve the semantic understanding and detection performance of the model (Jiang et al., 2023). Specifically, researchers use learnable vectors to construct soft prompts. These prompts are embedded into the text or entity information encoding process (Wei et al., 2025), enhancing the model's ability to capture textual semantics and entity relationships. Other works integrate prompt tuning with contrastive learning (Wang et al., 2025b). This combination helps reduce the gap between pretrained models and downstream tasks, improving adaptability and robustness. In addition, prefix tuning-based methods typically add task-specific prefixes before news content, guiding the model to focus on key information relevant to classification (Abdel-Salam, 2023).

However, these methods generally face the following limitations. First, prompts are usually designed separately for different modalities, and the prompt information is independent of the news content, failing to consider the contextual semantic relevance between prompts and content. Second, during multimodal feature fusion, most existing methods rely on simple concatenation or lightweight attention mechanisms. This leads to shallow interaction between modalities, and limits the model's ability to capture deep cross-modal semantic correlations. Moreover, many approaches focus primarily on learning global features, while neglecting the complementary relationship between local and global features. This restricts the model's comprehensive understanding of visual and textual information.

To address these limitations, we propose a multimodal shared prompt fake news detection method based on the pretrained model framework. Our method integrates task-related prompts with the semantic content of news, and uses a shared prompting mechanism to guide the pretrained model in jointly modeling multimodal features, thereby obtaining deeply interacted multimodal features. Then, to learn both coarse-grained and fine-grained features of the deep image-text hybrid information, we introduce a dual-perspective pseudo-siamese attention network. Finally, a classifier is used to classify the learning results of the dual-perspective pseudo-siamese attention network.

3. Research objectives

Our research objectives are to fully exploit the architectural advantages of the pretrained model. This aims to achieve efficient and accurate fake news detection, while minimizing computational resource consumption and time consumption. In this work, the specific research objectives include the following two points.

RO1. Multimodal shared prompt for pretrained models. Based on the complementarity of text and image information, we construct a multimodal shared prompt for pretrained text-image encoder. Our strategy introduces shared prompts to the encoder, facilitating deep interaction between visual and textual information. This allows the proposed model to leverage its inherent architectural strengths. It can maximize the extraction of deep correlations and complementarities between images and text, significantly enhancing the accuracy of fake news detection.

RO2. Dual-perspective pseudo-siamese attention. The multimodal interaction features contain both useful information and noise, effectively extracting the relevant features presents a key challenge. We introduce a dual-perspective pseudo-siamese attention network, which processes multimodal interaction features from both global and local perspectives in parallel. This dual-perspective approach, guided by global information, enables precise focus on crucial areas, thereby enhancing the model's ability to detect fake news accurately.

4. Proposed method

We use ViT and XLM-R as the foundational framework for the image encoder and text encoder (Conneau et al., 2020; Radford et al., 2021), and based on this, we propose a multimodal shared prompt and a dual-perspective pseudo-siamese attention network. The flowchart of DMSP is shown in Fig. 1. To better explain our method, we will now describe the encoders used in this study.

4.1. Image encoder and text encoder

We use ViT as the image encoder V , which consists of k transformer encoder layers, denoted $\{V_i\}_{i=1}^k$. Upon receiving an image, the encoder first divides the image I ($I \in \mathbb{R}^{H \times W \times 3}$, H and W are the height and width of the image) into equal-sized m patches, and uses a linear transformation to convert these patches into an embedding vector E_0 ($E_0 \in \mathbb{R}^{m \times d_V}$, d_V is the hidden dimension of V). A learnable class token c_i is then added to the head of the embedding vector E_0 . The embedding vector E_i is subsequently fed into the $(i + 1)$ th transformer encoder layer. The above process is represented as follows:

$$[c_i, E_i] = V_i([c_{i-1}, E_{i-1}]), i = 1, 2, 3, \dots, k. \quad (1)$$

To obtain the final image feature representation, the class token from the last transformer layer needs to be mapped to the encoding space through ImageProj. This process is shown below.

$$v = \text{ImageProj}(c_k), \quad (2)$$

where v represents the output of the image encoder and $v \in \mathbb{R}^{d_V}$, ImageProj is a linear projection layer.

We use XLM-R as the text encoder $\mathcal{L}$. It first tokenizes and marks the input text T , followed by word embedding $W_0 = [w_0^1, w_0^2, \dots, w_0^n]$, $W_0 \in \mathbb{R}^{n \times d_{\mathcal{L}}}$, w_0^n is the n th word embedding, $d_{\mathcal{L}}$ is the hidden dimension of $\mathcal{L}$. Then, the word embedding results W_i are input into the $(i + 1)$ th transformer layer of the text encoding branch $\mathcal{L}_{i+1}$. The final embedding representation that considers the entire text context is as follows:

$$t = [W_i] = \mathcal{L}_i(W_{i-1}), i = 1, 2, \dots, j, \quad (3)$$

where t is the output of the text encoder and $t \in \mathbb{R}^{d_{\mathcal{L}}}$, j is the number of transformer layer in the text encoding branch.

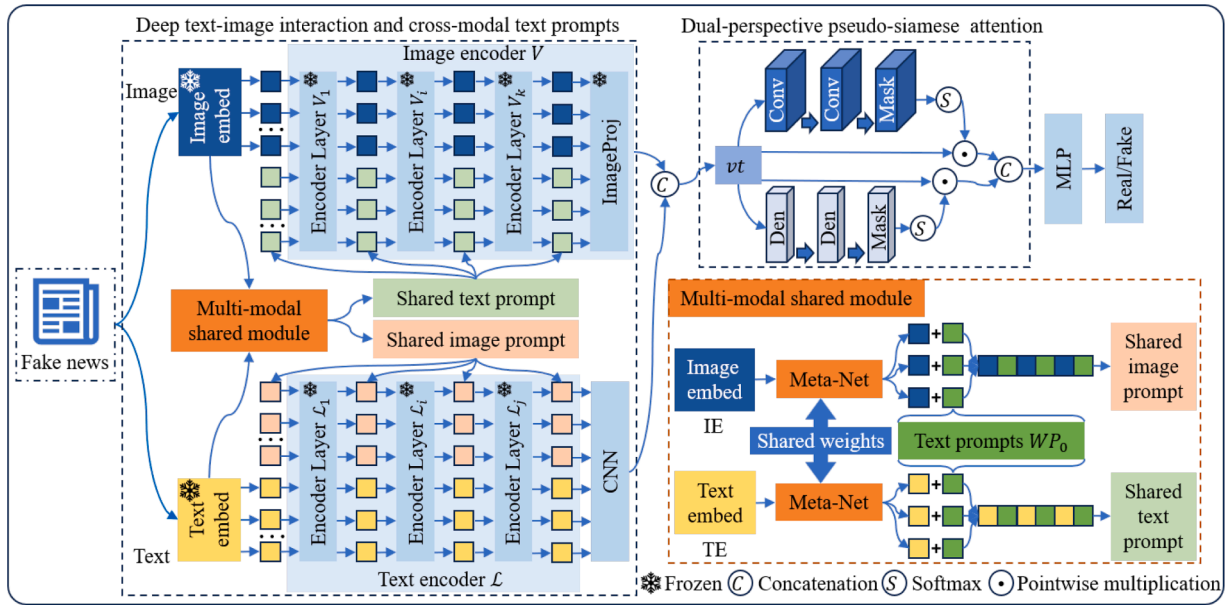


Fig. 1. Flowchart of DMSP. DMSP first generates shared image and text prompt information based on Meta-Net and text prompts, and then injects them into the text and image pretrained encoders, respectively. By fully leveraging the deep architecture of the pretrained encoders, DMSP obtains deeply interacted multimodal representations. Subsequently, DMSP employs a dual-perspective pseudo-siamese attention mechanism to perform parallel modeling of the multimodal information, capturing multi-scale features from both global and local perspectives.

4.2. Multimodal shared prompt

We propose a multimodal shared prompt learning approach that introduces task-oriented semantic guidance, enhancing the cross-modal alignment and interaction between visual and textual information. Unlike traditional methods that treat pretrained models as unimodal feature extractors, our shared prompt method fully taps into their interactive potential. It provides fine-grained semantic guidance for image-text interaction, and helps the model focus on key cues that are strongly linked to news authenticity. This approach benefits from the guidance of text prompts, enhancing the alignment ability between visual and textual information, and effectively guiding the model to capture task-specific semantic information (Zhou et al., 2022a,b).

Specifically, we first construct a learnable multimodal shared prompt using image encoding and text encoding. We then introduce shared prompts into both the pretrained image encoder and the pretrained text encoder, obtaining deep visual-textual interaction information from the text and image encoders. It is important to emphasize that the encoder parameters are frozen during training, ensuring that all image-text representations rely solely on the pretrained models themselves. We marked the frozen network layers with a snowflake symbol in Fig. 1. This process is described below. We begin by using the text encoder to tokenize and embed the news text, obtaining the embedded text information W_0 , while simultaneously using the image encoder to divide the image into patches and embed them, resulting in the image embedding vector E_0 . Then, both W_0 and E_0 are passed into the multimodal shared prompt module (MSM) to generate multimodal shared prompt, which combines image and text information. As shown in Fig. 1, MSM consists of a Meta-Net with shared weights and text prompts. MSM simultaneously takes text embedding and image embedding inputs, and uses the weight-shared Meta-Net to process the image and text embeddings in parallel, generating the final outputs. The Meta-Net is a spindly network structure made up of multiple linear layers. Next, a learnable text prompt WP_0 is used to guide and prompt the text features and the image features output by Meta-Net in parallel. This generates the shared text prompt and the shared image prompt after prompting. The above process is represented as follows.

$$STP_0, SIP_0 = \text{MSM}\{\text{Meta-Net}(TE, IE), WP_0\}, \quad (4)$$

where STP_0 is the shared text prompt, SIP_0 is the shared image prompt, WP_0 represents the encoded text prompt, TE is text embedding, IE is image embedding. The text prompt template used is [A NEWS OF REAL]. It is important to note that the text prompt template mentioned here is not the only one used in our study. We evaluated the proposed model using several different prompt templates, and a detailed discussion of these results is provided in Section 5.4.7. We then introduce the shared text prompt SIP_0 into the image encoder and the shared image prompt STP_0 into the text encoder, thus obtaining the deep mixed dual-stream image-text embedding vectors constructed by the image and text encoders. The process of introducing the multimodal shared text prompt vector into the image encoder can be represented as:

$$[c_i, E_i, STP_i] = V_i([c_{i-1}, E_{i-1}, STP_{i-1}]), i = 1, 2, \dots, k. \quad (5)$$

Finally, the output feature of the image encoder after applying multimodal shared prompt is denoted as vp , it can be expressed as:

$$vp = \text{ImageProj}([c_i, E_i, STP_i]). \quad (6)$$

The process of introducing the multimodal shared image prompt vector into the text encoder can be expressed as:

$$[W_i, SIP_i] = \mathcal{L}_i(W_{i-1}, SIP_{i-1}), i = 1, 2, \dots, j. \quad (7)$$

After applying the multimodal shared image prompt, the output feature of the text encoder is denoted as $tp = [W_i, SIP_i]$. The fused multimodal feature vt is obtained by concatenating the output feature of the text encoder and the output feature of the image encoder.

$$vt = \text{cat}(vp, tp), \quad (8)$$

where cat represents the concatenation operation.

4.3. Dual-perspective pseudo-siamese attention network

The fused multimodal feature often contains rich global associative features as well as critical local features. However, existing methods rarely perform further extraction of these global and local features, and instead directly use the fused multimodal feature as the final classification representation. We believe that deeply mining both global and local features can provide the model with more stable multi-scale representations, thereby enhancing its detection performance. Thus, to extract global associative features and critical local features from the multimodal feature vector vt , we propose a dual-perspective pseudo-siamese attention network. The proposed attention network consists of a fine-grained learning branch and a coarse-grained learning branch, aiming to model fused multimodal feature from both local and global perspectives. Unlike the standard attention that typically captures feature information at a single level, our dual-perspective pseudo-siamese attention uses two learning branches to focus on features at different levels. The fine-grained learning branch can provide the model with local features of fused multimodal information, while the coarse-grained learning branch can provide the model with global features of fused multimodal information. This collaborative local-global attention learns multimodal information from both coarse and fine perspectives, enhancing the model's ability to perceive key feature areas. The structure of the dual-perspective pseudo-siamese attention network is demonstrated in Fig. 1, the fine-grained and coarse-grained branch structures have the same number of network layers.

Fine-grained learning from local perspective. The first perspective focuses on fine-grained learning from a local perspective, performing local attention-guided fine-grained feature learning on the deep mixed features. This perspective begins by employing a dual-layer convolutional network to extract local features from the deep mixed information, enhancing non-linear expressiveness using the ReLU activation function. The masking operation is then applied to eliminate certain ineffective feature information, ensuring that only key local features are retained. Finally, a Softmax activation function is applied, and the result is multiplied by the original input vector to obtain the local attention-weighted features. The mathematical expression for this perspective is as follows:

$$lp = \text{Softmax}(\text{Masked}(\mathcal{R}(\text{Conv}(\mathcal{R}(\text{Conv}(vt)))))) \odot vt, \quad (9)$$

where lp is the result of fine-grained learning from local perspective. Conv denotes a convolution operation, we refer to the parameter configuration of the backbone convolutional network in MCAN (Wu et al., 2021) and set the size of the convolution kernel to 3, thereby ensuring the effectiveness of the convolution module. $\mathcal{R}$ denotes the ReLU activation function, Masked denotes the masking operation, and $\odot$ denotes the point-wise multiplication.

Coarse-grained learning from global perspective. The second perspective emphasizes coarse-grained learning from a global perspective, performing global attention-guided coarse-grained feature learning on the deep mixed information. In the dense network, each layer's neurons are fully connected to all neurons in the preceding layer, thereby receiving the complete set of feature information from the previous layer (Kaliyar et al., 2021). This design enables each layer to fully leverage information from earlier layers, effectively capturing global features (Narayanan, 2023). Thus, the second perspective begins with a dual-layer dense network to linearly transform the input features, followed by the application of the ReLU activation function for non-linearity. The masking operation is subsequently used to filter out ineffective feature information, preserving the most representative global features. Finally, an activation function is applied, and the result is multiplied by the original input vector to obtain the global attention-weighted features. The mathematical expression for this perspective is as follows:

$$gp = \text{Softmax}(\text{Masked}(\mathcal{R}(\text{Den}(\mathcal{R}(\text{Den}(vt)))))) \odot vt, \quad (10)$$

where gp is the result of coarse-grained learning from global perspective, Den denotes dense layer. After obtaining the local and global attention-weighted features, the dual-perspective joint output is achieved through a concatenation operation, represented as follows:

$$R = \text{cat}(lp, gp), \quad (11)$$

where R represents the output of the dual-perspective pseudo-siamese attention network.

Table 1
Dataset details.

News	PolitiFact			GossipCop		
	Fake	Real	Total	Fake	Real	Total
with textual information	420	528	948	4947	16694	21641
with visual information	336	447	783	1650	16767	18417

4.4. Prediction

We use a multi-layer perceptron (MLP) network with a Sigmoid output layer as the classifier. We adopt the MLP architecture in MDFEND (Nan et al., 2021) and set the number of MLP layers to 5, which has shown good performance in similar classification tasks. This network architecture systematically extracts features through multiple hidden layers. Finally, a Sigmoid activation function is applied to the output layer, compressing the model's output to a range between 0 and 1. This process can be expressed as:

$$\hat{y} = \text{Sigmoid}(\text{MLP}(R)), \quad (12)$$

where $\hat{y}$ is model's output value. Finally, we use binary cross-entropy (BCE) as the classification loss. This loss measures the difference between the model's predictions and the true labels, and guides the optimization process.

$$\mathcal{L}_{cls} = - \sum_{i=1}^N (y^i \log \hat{y}^i + (1 - y^i) \log(1 - \hat{y}^i)), \quad (13)$$

where y^i is i th actual value, $\hat{y}^i$ is i th model's output value.

5. Experiments

5.1. Experimental setups

We use the PyTorch library to train and optimize our model on a single GPU (NVIDIA GeForce RTX 4060 Ti 16GB). We search for the optimal learning rate within the range of $1e-2$ to $1e-6$. Ultimately, we set the learning rate to $5e-3$ for the PolitiFact dataset and $3.5e-6$ for the GossipCop dataset. Apart from the learning rate, we employ the same hyperparameters for training both datasets. Specifically, the batch size is 64, the random seed is 2021, the weight decay is $1e-5$, the number of epochs is 5, the maximum text length is 256, the dropout of MLP is 0.1, the text encoder pretrained model is xlmr-base, and the image encoder pretrained model is ViT-B/32. We use the default betas parameters (0.9, 0.99) for the Adam optimizer to train our model.

5.2. Datasets

We use the publicly available PolitiFact and GossipCop datasets from FakeNewsNet (Shu et al., 2020) to train and test the proposed algorithm. The PolitiFact dataset is collected from the well-known nonprofit political statement and report fact-checking website politifact.com, while the GossipCop dataset is collected from the fact-checking website gossipcop.com, which focuses on celebrity news and entertainment stories published in magazines and newspapers (Zhang et al., 2024). The truth labels of both datasets are provided by domain experts, ensuring the validity and authority of the data labels (Zhou et al., 2020). The statistics for real and fake news in these two datasets are shown in Table 1. Regarding the potential issue of missing modalities in multimodal samples, we followed the data preprocessing strategy in SAFE (Zhou et al., 2020), retaining only samples that contain both text and image modalities for model training and testing. And we adopt the same dataset splitting strategy as the baseline model TARD (Yang et al., 2024b), dividing the dataset into training and test sets in a 4:1 ratio.

5.3. Baselines

To verify the performance of the proposed method, we adopt state-of-the-art baseline models for comparison. Below is a brief introduction to the baseline models used.

- **MRE-FND (2024)** (Han et al., 2024), which improves the accuracy and interpretability of the detection model by integrating multi-source information, such as text, images, user data, related posts, topic information and knowledge graphs.
- **GAMC (2024)** (Yin et al., 2024), which introduces self-supervised learning to eliminate reliance on labeled data. The proposed graph auto-encoder combines reconstruction loss and contrastive loss to better capture latent features of propagation graphs.
- **PSGT (2024)** (Zhu et al., 2024), which designs a noise-reducing self-attention mechanism based on the information bottleneck principle. It can capture long-sequence dependencies and propagation structural in news dissemination, demonstrating broad application potential in real-world social media fake news detection task.
- **TARD (2024)** (Yang et al., 2024b), which treats comments in topic propagation processes as segments of audio signals, with the entire topic propagation process regarded as a piece of audio. By quantifying the propagation features, it maps them to the amplitude, frequency, and offsets of the audio signals.

Table 2
Outcome of model comparisons on PolitiFact and GossipCop datasets.

Method	PolitiFact				GossipCop			
	Acc	Pre	Rec	F1	Acc	Pre	Rec	F1
MRE-FND	0.929	0.931	0.947	0.939	0.834	0.873	0.918	0.894
GAMC	0.838	0.836	0.827	0.831	0.946	0.941	0.946	0.943
PSGT	0.917	0.966	0.918	0.922	0.980	0.874	0.987	0.980
TARD	0.915	0.921	0.935	0.928	0.943	0.947	0.977	0.962
SheepDog	0.911	0.912	0.911	0.911	0.762	0.763	0.762	0.762
MD-PCC	0.639	0.638	0.636	0.636	0.855	0.820	0.782	0.798
MPL	0.923	0.899	0.915	0.907	0.869	0.800	0.756	0.773
CFFN	0.814	0.816	0.805	0.808	0.925	0.908	0.871	0.888
PASTEL	0.771	0.774	0.753	0.771	0.686	0.408	0.747	0.646
RPPG-Fake	0.924	0.925	0.902	0.914	0.891	0.911	0.940	0.925
DGA-Fake	0.926	0.927	0.904	0.915	0.882	0.913	0.927	0.920
Proposed	0.966	0.971	0.960	0.964	0.990	0.987	0.983	0.985

- **SheepDog (2024)** (Wu et al., 2024), which adopts diverse prompts to guide large language model in reconstructing news content, generating stylistically varied news versions, adversarial test samples and content-relevant veracity attributions.
- **MD-PCC (2025)** (Wang et al., 2025a), which employs manual-crafted natural language prompt templates and in-context examples to guide the model, thereby extracting potentially contradictory common sense triples to enhance the input text.
- **MPL (2024)** (Hu et al., 2024), which develops a learnable prompt module that can adaptively generate high-quality prompt vectors, thereby enhancing the semantic modeling capabilities.
- **CFFN (2024)** (Li et al., 2024), which calculates the correlation scores between word-region pairs, categorizing them into high and low correlation parts. These parts are processed separately for consistency exploration and inconsistency calculation, and the results are selected cue to determine the credibility of multimodal news.
- **PASTEL (2024)** (Leite et al., 2024), which uses 19 complex confidence signals extracted by LLM for joint identification, achieving results close to supervised learning performance without the need for manual labeling.
- **RPPG-Fake (2025)** (Zhang et al., 2025), which constructs a heterogeneous graph composed of nodes such as publishers, news articles, and users. It designs a self-supervised dual-view contrastive learning mechanism, including node-view contrastive learning and meta-path-view contrastive learning, thereby capturing propagation topology features and heterogeneous relationship features.
- **DGA-Fake (2024)** (Zhang et al., 2024), which generates news propagation paths based on news content to provide rich social context information, and further extracts the global structural and temporal depth features of the propagation paths, improving the performance of fake news detection.

5.4. Performance analysis

5.4.1. Overall performance

In this subsection, we compare the performance of the proposed method with that of the baseline models, providing a detailed discussion and analysis of the results. For a fair comparison, we use the official metrics from the baseline models as the evaluation metrics, including accuracy (Acc), precision (Pre), recall (Rec), and F1 score (F1). The comparative experimental results are presented in Table 2.

The data in Table 2 indicate that our method demonstrates satisfactory performance on both public datasets, achieving detection accuracies of 96.6% and 99.0% on the PolitiFact and GossipCop datasets, respectively. This represents an improvement of 3.7% and 1% over the latest baseline models. To more intuitively present the performance of our method and baseline models across different datasets, we plotted the Acc and F1 histograms as shown in Fig. 2. From the figure, we observe that our method achieves a more balanced detection performance across different datasets, with minimal performance variation. Combining the data in the table, it can be seen that the accuracy difference between the proposed method and baseline models is only 2.4%. The difference for the TARD method is 2.8%, for RPPG-Fake it is 3.3%, for DGA-Fake it is 4.4%, for MPL it is 5.4%, for PSGT it is 6.3% and for the other baseline models, the differences exceed 8%. Although TARD and the proposed method exhibit similar performance differences, our method outperforms TARD by 5.1% and 5.7% in accuracy on the PolitiFact and GossipCop datasets, respectively.

Due to the powerful image-text encoding and feature extraction capabilities of the pretrained models, all the baseline models, except those based on large language models, are multimodal methods. These models mostly use pretrained models to extract image-text features, followed by the design of specific image-text interaction modules to capture the relationships between modality features. For instance, CFFN uses the BERT pretrained model for text encoding, then explores the correlations between text and image features. MRE-FND, on the other hand, uses the pretrained CLIP model for image-text encoding, and extracts relevant key information from the constructed knowledge graph. Clearly, these pretrained model-based methods have shown good detection performance on public datasets. SheepDog and PASTEL are LLM-based methods that rely primarily on the LLM to understand the textual context, while neglecting the content of news images. This results in relatively lower performance on the GossipCop dataset compared to the multimodal models. MD-PCC adds K in-context examples to hand-designed natural language prompt templates, which helps the model

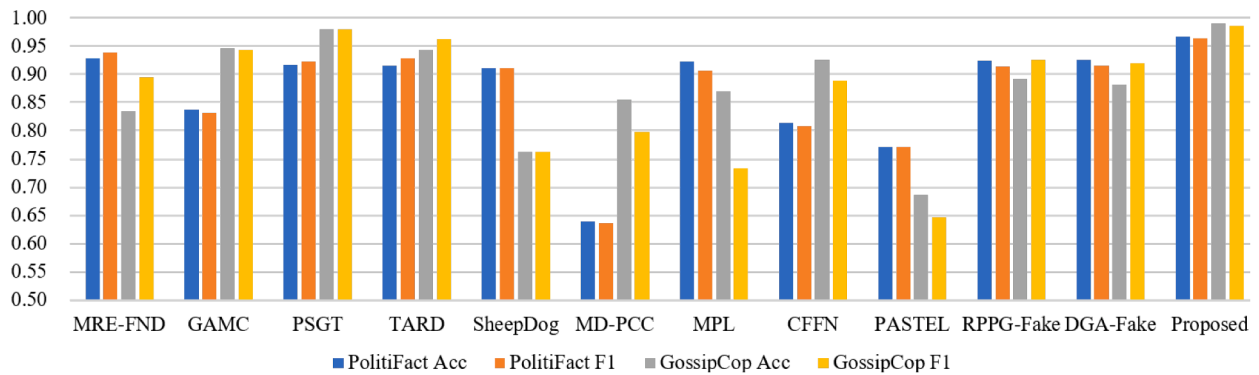


Fig. 2. Acc and F1 visualization on PolitiFact and GossipCop datasets.

better understand how to extract potentially conflicting triplets from news text. The extracted triplets are used to construct enhanced text, but the performance of this method is relatively weak on datasets with smaller data sizes. MPL uses the image and text encoders of the CLIP pre-trained model to extract image features and text features, and introduces a learnable prompt vector to calculate the similarity of the mixed multimodal features, thereby achieving good detection results on both datasets. PSGT and GAMC combine BERT encoding with graph nodes to effectively recognize fake news. GAMC improves detection accuracy by capturing latent features of the news propagation graph, while PSGT enhances detection accuracy by capturing long-sequence dependencies and propagation structure features in news dissemination. TARD transforms the topic space into an audio-like signal, mapping it to amplitude, frequency, and offset, achieving robust detection performance. RPPG-Fake enhances the incorporation of social context information by generating propagation paths, thereby improving the identification performance of fake news. DGA-Fake combines generative models with propagation path analysis to provide richer contextual information, which helps improve detection performance even in scenarios with limited data. Our method, in contrast to existing methods, employs multimodal shared prompt to achieve deep interaction between image and text information. Performance is further enhanced by coarse and fine-grained dual-perspective pseudo-siamese attention network, ultimately leading to performance that surpasses that of the baseline models.

5.4.2. Acc-Epoch scatter plot

To demonstrate the learning efficiency of the proposed model compared to baseline models across different datasets, we plotted the Acc-Epoch scatter plot in Fig. 3 (all epoch data correspond to the number of training epochs in the baseline models' experimental setup). It is evident that the proposed algorithm and SheepDog require the fewest training epochs, demonstrating satisfactory learning efficiency. This indicates that both algorithms have an advantage in training efficiency. Specifically, both algorithms achieve satisfactory detection performance after just five training epochs, and our method shows higher detection accuracy.

TARD and CFFN exhibit similar learning efficiency, requiring 41 and 50 epochs, respectively, to reach optimal detection accuracy. Although both algorithms perform well in terms of learning efficiency, they still require more training epochs compared to the proposed method and SheepDog. We speculate that these methods may not fully leverage the advanced architecture of pretrained models during training. GAMC requires 80 epochs to achieve optimal detection accuracy. Although the accuracy of GAMC is acceptable, its longer training time could lead to greater time and computational costs. PSGT shows higher detection accuracy on both the PolitiFact and GossipCop datasets, but requires 600 training epochs. PASTEL requires 500 epochs to reach its optimal accuracy, which indicates a significant time overhead during training, and it is difficult to achieve efficient learning within a short training time. We believe the reason why our method achieves optimal performance within five training epochs is as follows. First, our method is built on the pretrained vision-language model CLIP, which possesses strong semantic feature extraction and cross-modal alignment capabilities. This provides the model with robust and high-quality visual and textual features, significantly reducing the burden of learning effective representations during training and accelerating convergence. This is consistent with the baseline model SheepDog, which also utilizes a pretrained large language model and achieves optimal performance within just five epochs. Second, during the training process, we froze the parameters of the encoders, effectively reducing the number of trainable parameters and the difficulty of model optimization, thereby improving training efficiency. In summary, the proposed algorithm and SheepDog benefit from their foundational use of pretrained models, allowing them to achieve optimal detection accuracy with fewer training epochs. Other models, although competitive in accuracy, require longer training times, which may lead to higher computational costs and time consumption in practical applications. Therefore, the advantages of our method in efficiency and performance provide strong support for its use in real-world applications.

5.4.3. t-SNE visualization

t-SNE is a powerful data visualization tool. It maps high-dimensional data to a low-dimensional space, visually presenting the clustering structure of the data. When the model effectively separates data points from different categories in the high-dimensional feature space, the t-SNE results will show clear clusters with distinct inter-class gaps. This indicates that the model has strong classification ability. On the other hand, if the t-SNE plot shows considerable overlap between different categories, it suggests that the

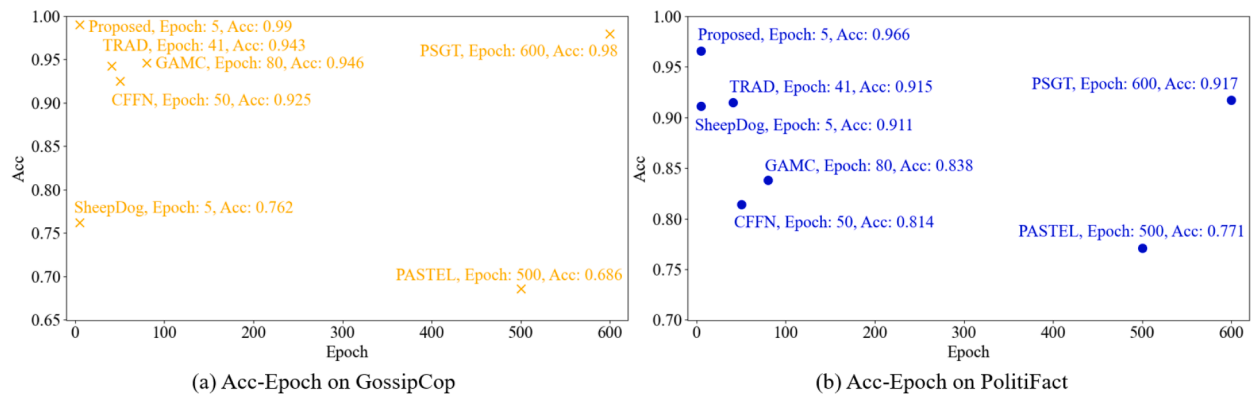


Fig. 3. Acc-Epoch with different models.

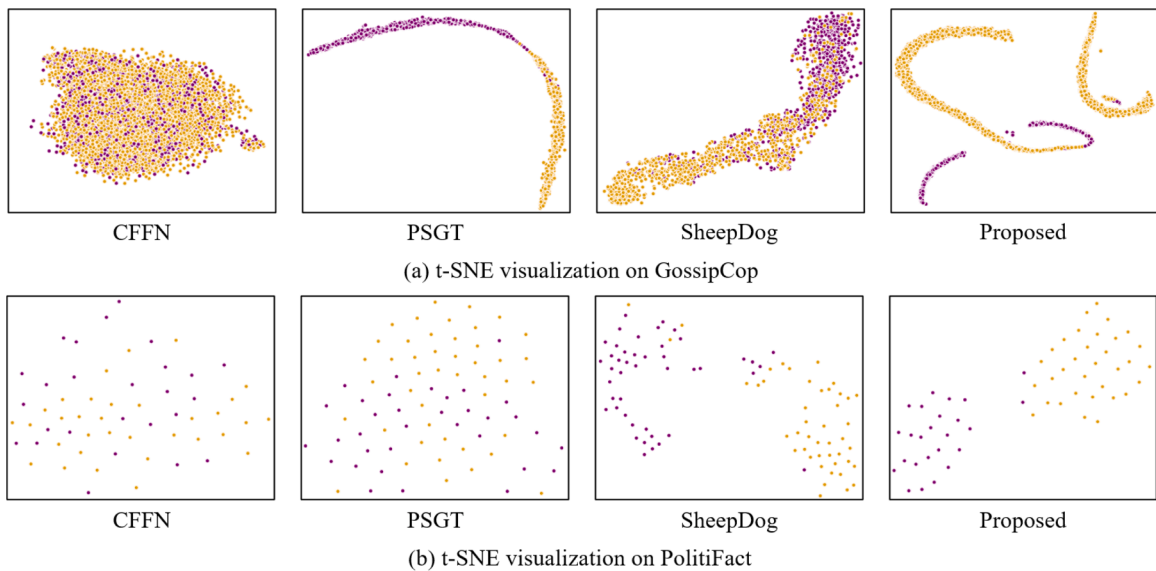


Fig. 4. t-SNE visualization on GossipCop and PolitiFact datasets. The yellow and purple points represent the fake news feature and real news feature after dimensionality reduction, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

model's ability to distinguish between these categories is insufficient. In this section, we use t-SNE to further validate the performance of the DMSP and the baseline models, and the results are shown in Fig. 4.

In this figure, yellow points represent the fake news feature points after the dimensionality reduction, while purple points represent the real news feature points. As shown in Fig. 4, our method exhibits clear clustering effects on both public datasets, with well-defined gaps between clusters and minimal overlap between different categories. PSGT and SheepDog algorithms also show good classification performance, but SheepDog has more overlap in clusters on the GossipCop dataset compared to PSGT. On the PolitiFact dataset, SheepDog shows better clustering with less overlap. PSGT and CFFN do not demonstrate good clustering results on the PolitiFact dataset. Additionally, CFFN does not show an effective data clustering capability on the GossipCop dataset, with substantial overlap between different categories.

5.4.4. Statistical validation

To further verify reliability and effectiveness, we conducted comprehensive statistical validation through multiple independent experiments and statistical significance testing.

Performance with different random seeds. Specifically, to eliminate the potential randomness of the experimental results, we performed eight independent experiments with different random seeds. The corresponding experimental results are presented in Table 3. As shown in this table, the proposed method achieves an average detection performance of 0.968 and 0.991 under different random seeds. Taking average accuracy as a benchmark, the maximum fluctuation range in the PolitiFact dataset is -0.02 to +0.015, while in the GossipCop dataset, the fluctuation range is -0.007 to +0.005. We speculate that the larger fluctuation in accuracy on

Table 3
Performance of DMSP with different random seeds.

Random seed	24	321	1249	1520	1556	2021	3224	3469
Acc on PolitiFact	0.983	0.966	0.966	0.948	0.983	0.966	0.966	0.966
Average Acc				0.968				
Acc on GossipCop	0.987	0.984	0.995	0.993	0.994	0.990	0.996	0.992
Average Acc				0.991				

Table 4
Outcome of *t*-test between DMSP and baselines.

Method	MRE-FND	GAMC	PSGT	TARD	SheepDog	CFFN	PASTEL	RPPG-Fake	DGA-Fake
p-value on PolitiFact	2.3e-05	6.1e-09	3.8e-06	3.0e-06	1.8e-06	1.9e-09	3.4e-10	1.0e-05	1.4e-05
p-value on GossipCop	1.6e-12	9.4e-09	1.0e-04	6.0e-09	1.1e-13	6.6e-10	1.6e-16	3.7e-11	2.0e-11

Table 5
Computational resource consumption.

Method	PolitiFact					GossipCop				
	CFFN	PSGT	SheepDog	PASTEL	Proposed	CFFN	PSGT	SheepDog	PASTEL	Proposed
Total training time (s)	1588	1674	2305	4466	69	15314	13564	38364	120146	3127
Average GPU usage (MB)	6292	5441	6800	5558	3447	5854	3561	6123	5425	3461
GFLOPs	8.77	0.16	43.53	909.06	157.13	8.77	0.16	43.53	909.06	157.13
Param (M)	114.96	0.56	85.65	262.41	265.19	114.96	0.56	85.65	262.41	265.19

PolitiFact is mainly caused by its smaller dataset size, which leads to fewer robust features being learned by the model. In contrast, with the larger data volume of GossipCop, the model is able to learn more robust features, significantly reducing the variation in detection accuracy. Overall, the results indicate that our method exhibits satisfactory stability and robustness across both datasets.

Statistical significance testing. Moreover, to validate the statistical significance of performance gains, we performed a statistical significance testing (*t*-test) between our method and the baseline models. The results are shown in Table 4. It can be observed that all p-values on the two public datasets are less than 0.05, indicating that the performance improvements of the DMSP over the baselines are statistically significant. In summary, the proposed approach not only outperforms existing baseline models, but also shows that its superior performance is statistically reliable and not due to chance.

5.4.5. Training time and computational resource

To further validate the advantages of our method in terms of computational resources and time consumption, we compared the GPU resource usage and the total training time of our method with the baseline models. The statistical results are shown in Table 5. From the data presented in the table, it can be observed that our method achieves the lowest total training time and GPU resource consumption on both datasets. On the PolitiFact dataset, the GPU usage of the proposed method is at least about 1990 MB lower than that of the baseline models, and the total training time is much shorter than that of the baseline model. On the GossipCop dataset, although our method has GPU resource usage comparable to PSGT, its total training time is only about one-fourth of PSGT's, demonstrating a significant improvement in training efficiency. We believe these advantages mainly stem from the fact that our approach fully exploits pretrained model. By introducing shared prompts and freezing pretrained parameters, we avoid the heavy burden of additional large-scale parameter training. In contrast, CFFN and PSGT require many more training epochs, leading to much longer training durations. SheepDog involves multiple complex iterations during actual training, which further prolongs the training cycle. Moreover, PASTEL has the longest training time among all models, due to its reliance on large language models to extract more than a dozen complex signals. Compared to PASTEL, our method has fewer GFLOPs (PASTEL is 909.06, proposed method is 157.13) and a comparable number of parameters. Although the proposed method has more parameters (265.19) and GFLOPs than lightweight models such as PSGT and CFFN, this reflects additional structural investment in multimodal feature interaction and iterative guided modeling. In comparison with SheepDog and PASTEL, the proposed method achieves better detection performance while maintaining a controllable computational scale. These results indicate that although the proposed method does not minimize parameter count or FLOPs, its training efficiency and memory consumption remain within acceptable limits, demonstrating a reasonable balance between performance gains and computational cost. In summary, our method requires significantly less training resources and time cost, but shows better performance than the baseline models.

5.4.6. Confusion matrix and AUC-ROC

Since the real and fake data in the GossipCop dataset are not balanced, we use the confusion matrix and AUC-ROC to further test the model performance. The test results are shown in Fig. 5. For all methods, the values on the diagonal of the confusion matrix are significantly higher than the misclassification counts on the off-diagonal. And our method consistently shows the best performance on both datasets, demonstrating a stronger discrimination ability. On the GossipCop dataset with a highly unbalanced

	CFFN	PSGT	PASTEL	SheepDog	Proposed																																																							
GossipCop	Confusion matrix <table><tr><td>Actual Label</td><td>Real</td><td>2405</td><td>82</td></tr><tr><td>Fake</td><td>159</td><td>559</td></tr><tr><td></td><td>Predicted Label</td><td>Real</td><td>Fake</td></tr></table>	Actual Label	Real	2405	82	Fake	159	559		Predicted Label	Real	Fake	Confusion matrix <table><tr><td>Actual Label</td><td>Real</td><td>663</td><td>20</td></tr><tr><td>Fake</td><td>9</td><td>674</td></tr><tr><td></td><td>Predicted Label</td><td>Real</td><td>Fake</td></tr></table>	Actual Label	Real	663	20	Fake	9	674		Predicted Label	Real	Fake	Confusion matrix <table><tr><td>Actual Label</td><td>Real</td><td>1077</td><td>499</td></tr><tr><td>Fake</td><td>135</td><td>349</td></tr><tr><td></td><td>Predicted Label</td><td>Real</td><td>Fake</td></tr></table>	Actual Label	Real	1077	499	Fake	135	349		Predicted Label	Real	Fake	Confusion matrix <table><tr><td>Actual Label</td><td>Real</td><td>642</td><td>150</td></tr><tr><td>Fake</td><td>244</td><td>548</td></tr><tr><td></td><td>Predicted Label</td><td>Real</td><td>Fake</td></tr></table>	Actual Label	Real	642	150	Fake	244	548		Predicted Label	Real	Fake	Confusion matrix <table><tr><td>Actual Label</td><td>Real</td><td>2475</td><td>12</td></tr><tr><td>Fake</td><td>21</td><td>697</td></tr><tr><td></td><td>Predicted Label</td><td>Real</td><td>Fake</td></tr></table>	Actual Label	Real	2475	12	Fake	21	697		Predicted Label	Real	Fake
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AUC-ROC	0.873	0.979	0.702	0.751	0.983																																																							
PolitiFact	Confusion matrix <table><tr><td>Actual Label</td><td>Real</td><td>19</td><td>7</td></tr><tr><td>Fake</td><td>4</td><td>29</td></tr><tr><td></td><td>Predicted Label</td><td>Real</td><td>Fake</td></tr></table>	Actual Label	Real	19	7	Fake	4	29		Predicted Label	Real	Fake	Confusion matrix <table><tr><td>Actual Label</td><td>Real</td><td>32</td><td>2</td></tr><tr><td>Fake</td><td>5</td><td>40</td></tr><tr><td></td><td>Predicted Label</td><td>Real</td><td>Fake</td></tr></table>	Actual Label	Real	32	2	Fake	5	40		Predicted Label	Real	Fake	Confusion matrix <table><tr><td>Actual Label</td><td>Real</td><td>30</td><td>11</td></tr><tr><td>Fake</td><td>8</td><td>31</td></tr><tr><td></td><td>Predicted Label</td><td>Real</td><td>Fake</td></tr></table>	Actual Label	Real	30	11	Fake	8	31		Predicted Label	Real	Fake	Confusion matrix <table><tr><td>Actual Label</td><td>Real</td><td>44</td><td>1</td></tr><tr><td>Fake</td><td>7</td><td>38</td></tr><tr><td></td><td>Predicted Label</td><td>Real</td><td>Fake</td></tr></table>	Actual Label	Real	44	1	Fake	7	38		Predicted Label	Real	Fake	Confusion matrix <table><tr><td>Actual Label</td><td>Real</td><td>23</td><td>2</td></tr><tr><td>Fake</td><td>0</td><td>33</td></tr><tr><td></td><td>Predicted Label</td><td>Real</td><td>Fake</td></tr></table>	Actual Label	Real	23	2	Fake	0	33		Predicted Label	Real	Fake
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AUC-ROC	0.805	0.915	0.763	0.911	0.960																																																							

Fig. 5. Confusion matrix and AUC-ROC on GossipCop and PolitiFact datasets.

data distribution, our method still exhibits satisfactory robustness and discriminative capacity, achieving an AUC-ROC score as high as 0.983, indicating a strong ability to distinguish between positive and negative samples. The corresponding confusion matrix shows that the number of misclassified samples is very small, with highly accurate predictions for both real and fake news. This further demonstrates the method's effectiveness under unbalanced condition. In contrast, although the PSGT method achieves an AUC-ROC value close to our method, it has relatively more misclassifications according to the confusion matrix. The CFNN method performs well in detecting real news, but suffers from significantly more misclassifications of fake news, leading to a lower AUC-ROC value. Additionally, both PASTEL and SheepDog have a large number of misclassifications, with AUC-ROC scores of 0.702 and 0.751, respectively, which is consistent with their confusion matrix results. This indicates that their performance is unstable when applied to unbalanced dataset. On the PolitiFact dataset, our method also shows satisfactory performance. The confusion matrix reveals virtually no notable misclassifications and the AUC-ROC reaches 0.960, ranking highest among all methods. SheepDog and PSGT perform similarly on this dataset, with AUC-ROC scores of 0.911 and 0.915, respectively, and a similar number of misclassifications. In contrast, CFNN and PASTEL perform relatively poorly, with AUC-ROC values of 0.805 and 0.763, and more misclassifications are shown in their confusion matrices. In summary, whether on the highly unbalanced GossipCop dataset or the smaller-scale PolitiFact dataset, our method achieves the best results in both the confusion matrix and AUC-ROC. This strongly validates the effectiveness, robustness, and advancement of the proposed method, and further demonstrates its applicability and superior performance in the multimodal fake news detection task.

5.4.7. Performance with different prompts

In this subsection, we use different prompts to verify the performance of the proposed method. We refer to the prompt design strategy presented in SAMPLE (Jiang et al., 2023), and conduct comprehensive experiments on our method using a total of ten different prompts. The experimental results are shown in Table 6. As can be seen from the data, despite differences in the expression of prompt sentences, our method consistently performs well on both datasets, indicating strong robustness to prompt variations and overall performance stability. On the PolitiFact dataset, the average accuracy of all prompts reaches 0.967, and the average F1 score is 0.964. On the GossipCop dataset, the average accuracy is 0.989, and the average F1 score is 0.984. These results demonstrate that our method maintains high discriminative power when using different prompts. We also observe that our method achieves better performance when using concise and complete natural language prompts, such as “This is real”. In contrast, performance decreases slightly when using prompts with incomplete language structures, such as “A real news”. Moreover, although the prompt “This is a piece of real news” is relatively long, the model still maintains good performance, indicating that our method is also adaptable to prompts of different lengths.

And it can be observed that prompts containing explicit “fake” semantics tend to degrade model performance to some extent. For example, “This is fake” yields accuracies of 0.948 and 0.977 on PolitiFact and GossipCop, respectively, both below the average performance. This suggests that explicitly injecting label-indicative semantics may interfere with the model's ability to fully exploit discriminative features. In contrast, neutral prompts lead to a slight performance improvement without introducing any prior bias toward real or fake labels. “This is news” achieves accuracies of 0.983 and 0.998 on the two datasets, both above the average, indicating that the proposed method does not rely on explicit “real” or “fake” semantics in the prompts for decision making, but instead primarily bases its predictions on image-text consistency modeling and cross-modal feature interaction. These results demonstrate that the proposed method is robust to prompt semantics, and that its performance gains are not driven by implicit label-semantic bias in the prompts, but rather by effective modeling and reasoning over multimodal.

Table 6
Performance of DMSP with different prompts.

Prompt	PolitiFact				GossipCop			
	Acc	Pre	Rec	F1	Acc	Pre	Rec	F1
This is real	0.983	0.985	0.980	0.982	0.997	0.998	0.992	0.995
A real news	0.948	0.958	0.940	0.946	0.986	0.978	0.981	0.979
A news of real	0.966	0.971	0.960	0.964	0.990	0.987	0.983	0.985
This news is real	0.966	0.971	0.960	0.964	0.995	0.995	0.991	0.993
This is real news	0.983	0.985	0.980	0.982	0.992	0.987	0.989	0.988
This is a piece of real news	0.966	0.963	0.970	0.965	0.973	0.965	0.955	0.960
This is fake	0.948	0.950	0.945	0.947	0.977	0.977	0.957	0.967
A news of fake	0.963	0.961	0.933	0.946	0.983	0.985	0.98	0.982
This is news	0.983	0.985	0.982	0.982	0.998	0.999	0.995	0.997
A news report	0.966	0.963	0.970	0.965	0.995	0.995	0.99	0.993
Average	0.967	0.969	0.962	0.964	0.989	0.987	0.981	0.984

Table 7
Ablation study results.

Ablation			PolitiFact				GossipCop			
Encoder	MSP	DPAN	Acc	Pre	Rec	F1	Acc	Pre	Rec	F1
✓			0.741	0.737	0.734	0.735	0.799	0.756	0.577	0.582
✓	✓		0.914	0.912	0.919	0.913	0.956	0.972	0.902	0.931
✓	✓	✓	0.966	0.971	0.960	0.964	0.990	0.987	0.983	0.985

Table 8
Misclassified case.

Dataset	News text	News image	Actual label	Prediction	Analysis
PolitiFact	@RandPaul: Will @realDonaldTrump betray voters by hiring John Bolton?		Real	Fake	Multiple names in the text cannot be aligned with a single person in the image
GossipCop	Kim Kardashian when in Africa you wear your passport like its an accessory. That's the last place you want to be w/o your passport.		Fake	Real	A single name in the text cannot be aligned with multiple people in the image

5.5. Ablation study

We perform ablation studies on two datasets, and the ablation results are shown in Table 7. In this table, Encoder refers to the image encoder and text encoder, MSP represents the multimodal shared prompt, and DPAN represents the dual-perspective pseudo-siamese attention network. The MSP results illustrate that after introducing learnable shared feature prompt vectors into the encoder, the detection accuracy shows a significant improvement on both datasets. Specifically, on the PolitiFact dataset, the accuracy increased from 74.1% to 91.4%, an improvement of 17.3%, and on the GossipCop dataset, the accuracy increased from 79.9% to 95.6%, an improvement of 15.7%. This demonstrates that the multimodal shared prompt provides highly effective multimodal mixed prompt information, allowing the model to obtain deep interaction feature vectors from both the image and text, greatly enhancing its detection performance. Further, after the dual-perspective pseudo-siamese attention network processes the mixed image-text features from both global and local perspectives, the model's detection performance is further improved. In the PolitiFact dataset, accuracy increased from 91.4% to 96.6%, an improvement of 5.2%, and in the GossipCop dataset, accuracy increased from 95.6% to 99.0%, an improvement of 3.4%. This indicates that the dual-perspective pseudo-siamese attention network provides the model with more refined and accurate features, leading to a further increase in detection accuracy.

5.6. Training loss

To further investigate the rapid convergence of the model within five epochs, we conducted a systematic tracking and analysis of the training loss trajectory, as shown in Fig. 6. The loss curves indicate that the proposed method exhibits fast and stable convergence on both datasets. On the GossipCop dataset, the training loss decreases monotonically from 0.675 in the first epoch to 0.049 in the fifth epoch without noticeable oscillations, suggesting that the model reaches a stable optimization state within a small number of training epochs. On the PolitiFact dataset, due to its relatively smaller sample size and sparser class distribution, a certain fluctuation

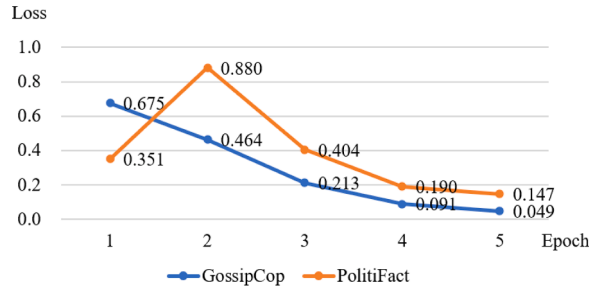


Fig. 6. Training loss on GossipCop and PolitiFact datasets.

Table 9
Comparisons with LMMs on PolitiFact and GossipCop datasets.

Method	GossipCop		PolitiFact		GFLOPs	Param (M)	Time (s)	GPU (GB)
	Acc	F1	Acc	F1				
LLaVA	0.836	0.729	0.672	0.649	4191.25	7063.43	0.499	14.52
Qwen2-VL	0.817	0.647	0.793	0.776	2007.75	2208.99	0.400	5.23
Deepseek-VL	0.902	0.847	0.845	0.844	1530.58	7343.99	0.579	15.62
Proposed	0.990	0.985	0.966	0.964	157.13	265.19	0.034	3.37

is observed at the second epoch. However, the loss quickly drops thereafter and decreases from an initial value of 0.351 to 0.147 by the fifth epoch, still showing a clear convergence trend overall.

From a mechanistic perspective, this training efficiency can be attributed to the combined effects of two factors. First, prompt learning constrains the optimization space to a low-dimensional and controllable set of prompt parameters, significantly reducing the difficulty of parameter search and enabling the model to rapidly align with the task decision boundary at early training stages. Second, the large-scale pretrained encoders remain frozen during training, and their parameter stability effectively prevents training oscillations caused by large-scale gradient updates, thereby further enhancing the stability and convergence speed of the overall optimization process. In summary, the loss trajectory analysis demonstrates that the fast convergence of the proposed method is not incidental, but rather jointly driven by the constrained optimization space and the frozen pretrained parameters, which explains why the model can achieve stable detection performance within only five training epochs.

5.7. Compared with LMMs

To compare our method with large-scale multimodal models (LMMs), we conducted systematic comparative experiments between our method and representative LMMs (LLaVA (Liu et al., 2023), Qwen2-VL (Wang et al., 2024b), and DeepSeek-VL (Lu et al., 2024)) on GossipCop and PolitiFact datasets.

To ensure a fair comparison, we employed the same classifier from our method for all models and kept the training strategies and hyperparameter settings strictly identical. The experimental results are reported in Table 9, where Time denotes the average inference time per sample, and GPU indicates the average GPU memory usage. The model checkpoints used for LLaVA, Qwen2-VL, and DeepSeek-VL are llava-1.5-7b-hf, Qwen2-VL-2B, and deepseek-vl-7b-base, respectively.

The results in Table 9 show that although LMMs exhibit strong general-purpose multimodal modeling capabilities, our method consistently outperforms these LMMs on the task of fake news detection across both GossipCop and PolitiFact. Moreover, our approach demonstrates clear differentiated competitiveness in terms of model complexity and inference efficiency. Specifically, both DeepSeek-VL and LLaVA are 7B-scale models, with parameter sizes of 7343.99 M and 7063.43 M, approximately 27 times and 26 times larger than our method. Their computational costs are also significantly higher, at 1530.58 GFLOPs and 4191.25 GFLOPs, respectively, while our model requires 157.13 GFLOPs. In terms of GPU memory consumption, DeepSeek-VL and LLaVA require 15.62 GB and 14.52 GB, respectively, whereas our method uses only 3.37 GB. Even Qwen2-VL, which has a relatively smaller parameter size (2208.99 M), still incurs much higher GFLOPs (2007.75) and GPU memory usage (5.23 GB) than our approach. Regarding inference efficiency, our method achieves an average inference time of only 0.034 s per sample, which is significantly faster than LLaVA (0.499 s), Qwen2-VL (0.400 s), and DeepSeek-VL (0.579 s). Overall, these results demonstrate that our method attains superior detection performance while maintaining substantially lower parameter counts, computational complexity, memory consumption, and inference latency, thereby highlighting its strong differentiated competitiveness compared to advanced LMMs.

5.8. Error case analysis

To identify the limitations of the proposed model and discover potential directions for improvement, we selected representative misclassified cases from both datasets for analysis, and the cases are listed in Table 8. In the PolitiFact dataset, we observed that when the text mentions multiple individuals but the image contains only one person, the model often makes incorrect predictions.

We speculate that this is due to the model struggling to match people in the image with all entities in the text, leading to misalignment at the entity level and affecting the final judgment of authenticity. Similarly, in the GossipCop dataset, when the text refers to only one individual, but the image contains multiple people, the model faces ambiguity in cross-modal alignment at the entity level. It struggles to identify which person in the image is most relevant to the text, resulting in inaccurate feature matching and misclassification.

These cases reveal that our model faces limitations when dealing with inconsistent entities between text and image. In the future, we plan to incorporate named entity recognition to explicitly align textual entities with visual individuals, thereby enhancing the model's discrimination ability and robustness in complex multi-entity scenarios.

5.9. Implications

This study explores the potential of combining prompt information with pretrained models for fake news detection, and proposes an efficient detection method.

At the theoretical level, this work demonstrates the feasibility and effectiveness of the shared prompt mechanism in pretrained models. By introducing multimodal shared prompt vectors into the model, we enable deeper cross-modal interaction and semantic guidance, which significantly enhances the model's ability to distinguish between real and fake information. This provides a new theoretical perspective for adapting multimodal pretrained models to downstream tasks. At the practical level, the proposed method improves detection performance while maintaining robustness and stability, and also reduces computational cost and training time. Experiments show that our method achieves strong results on public datasets and greatly lowers training overhead, confirming its applicability and practical value.

Overall, this study highlights the potential of integrating prompt information with pretrained models for fake news detection, providing useful insights for future work on prompt-driven multimodal understanding.

6. Conclusion

In this work, we propose a fake news detection method based on pretrained model, which incorporates a learnable multimodal shared prompt module to generate image-text shared prompts. These multimodal shared prompts are then introduced into the pretrained image-text model, leveraging the advanced architecture of the pretrained models to obtain dual-stream image-text mixed feature vectors. Subsequently, the dual-view pseudo-siamese attention mechanism is applied to the mixed image-text feature vectors, performing attention computation from both global and local perspectives to produce the final attention results. Comparative experiments demonstrate that our method is an effective fake news detection approach, achieving detection performance superior to advanced baseline models with just five training epochs. Ablation studies show that each of the proposed modules is effective, and makes a significant contribution to improving detection accuracy.

The main limitations of this study are as follows. First, there may be latent distribution biases across different datasets, such as differences in news topics, linguistic styles, and visual content, which may lead to inconsistent model performance. Second, although the proposed method performs well on political and entertainment datasets, domain shifts may still affect its generalization in cross-domain or cross-platform applications. In addition, although the pseudo-siamese structure shows strong capability in dual-view feature extraction, its parallel design introduces extra computational cost, which may limit scalability in resource-constrained environments.

In future research, we will introduce debiasing techniques to reduce dataset and domain discrepancies. At the same time, we will incorporate a mixture-of-experts mechanism to dynamically assign expert modules for different news categories, enabling adaptive modeling across various types of fake news. Furthermore, to address scenarios with limited data or imbalanced labels, we will explore few-shot and other low-resource learning strategies to enhance robustness and generalization under small-sample conditions. In addition, we will investigate lightweight model designs to reduce computational cost while maintaining accuracy, improving deployment efficiency and practical applicability. We also plan to introduce human coder validation and combine it with the automated validation, in order to supplement human-level contextual understanding and explore more practical detection strategies.

CRedit authorship contribution statement

Shouxin Liu: Writing – original draft, Methodology, Conceptualization; **Hongran Zeng:** Validation, Methodology; **Junfeng Guo:** Writing – original draft, Formal analysis; **Chongyang Zhang:** Writing – review & editing, Validation; **In-Kwon Lee:** Writing – review & editing, Data curation; **Yushu Zhang:** Writing – review & editing, Investigation; **Xiaowei Li:** Supervision, Resources.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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